

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Criteria	Weight age	Level 4 – Exemplar y	Level 3 – Proficient	Level 2 – Developin g	Level 1 – Beginnin g	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	

Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
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ASSESSMENT TOOL 3 – Hackathon

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5-AT3 Hackathon	Assessment:	ASSESSMENT TOOL 3 – Hackathon
Weightage:	30 % of CIA	Total Marks:	100
CO4 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	M V RAGHAVAN	Reg. No.:	192424373
Batch	2023	Date:	29-08-2026

Code of Conduct

I M V RAGHAVAN (192424373) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M V RAGHAVAN

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	

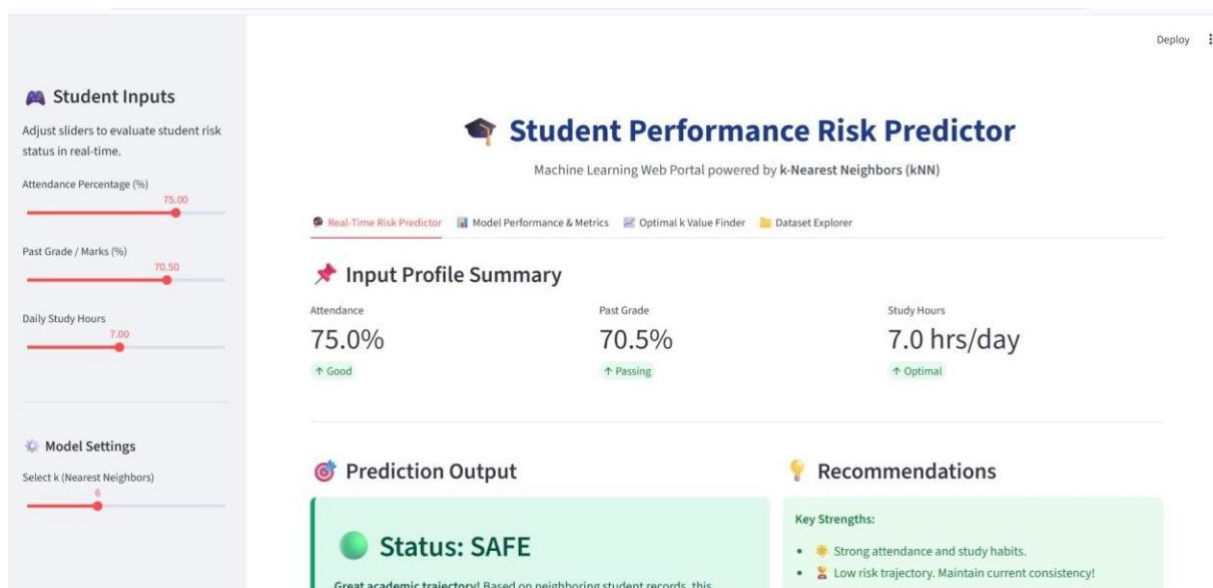
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	
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Problem Statement:

Students at academic risk are often identified too late for effective intervention. This system uses attendance, past grades, and study hours with the kNN algorithm to classify students as At Risk or Safe, helping educators identify at-risk students early and provide timely academic support.

Code:

```
E: > surveillance > student.py > ...
8
9  import sys
10 import subprocess
11
12 # -----
13 # STEP 0: AUTO-INSTALL MISSING LIBRARIES & AUTO-LAUNCH WEB APP
14 # -----
15 required_packages = {
16     "streamlit": "streamlit",
17     "pandas": "pandas",
18     "numpy": "numpy",
19     "sklearn": "scikit-learn",
20     "matplotlib": "matplotlib",
21     "seaborn": "seaborn"
22 }
23
24 for module_name, pip_name in required_packages.items():
25     try:
26         __import__(module_name)
27     except ImportError:
28         print(f"[INFO] Installing missing library: '{pip_name}'...")
29         subprocess.check_call([sys.executable, "-m", "pip", "install", pip_name])
30
31 import streamlit as st
32
33 # Auto-launch Streamlit web server if executed via standard 'python student.py'
34 if not st.runtime.exists():
35     print("\n" + "="*60)
36     print("LAUNCHING STUDENT RISK PREDICTOR WEBSITE IN YOUR BROWSER...")
37     print("="*60 + "\n")
38     subprocess.run([sys.executable, "-m", "streamlit", "run", sys.argv[0]])
39     sys.exit(0)
40
41 # Standard imports once inside Streamlit runtime
42 import os
43 import warnings
```



Student Inputs

Adjust sliders to evaluate student risk status in real-time.

Attendance Percentage (%)

75.00

Past Grade / Marks (%)

70.50

Daily Study Hours

7.00

Model Settings

Select k (Nearest Neighbors)

6

Real-Time Risk Predictor

Model Performance & Metrics

Optimal k Value Finder

Dataset Explorer

k Value vs. Test Accuracy

Tested $k \in [1, 15]$. Best Accuracy (96.7%) is achieved at $k = 13$.

